

PATCHED != NOT COMPROMISED

SOC Investigation Report - Simulated Defensive Lab

Track	DSOU / Blue Team
Mission	Post-patch compromise investigation
Analyst	Nimisha Sinha
Environment	Kali Linux VM + Windows host (Host-Only lab)
Method	Case 1 simulated suspicious evidence; Case 2 real normal SSH login

Objective: Determine whether suspicious activity existed before a security patch was applied. The lab intentionally compares a suspicious case against a clean/normal case using the same automated checker.

Important: No real vulnerability was exploited in this exercise. The "Security patch applied" event and suspicious indicators in Case 1 were simulated for defensive investigation practice.

Executive Result

Case 1 - Simulated suspicious activity	HIGH RISK	Risk Score 11
Case 2 - Normal Windows-to-Kali SSH activity	LOW RISK	Risk Score 0

1. Lab Setup and Evidence Generation

The test used a dedicated soc-incident-lab directory on Kali. Case 1 generated a mix of normal and suspicious journal events, then created harmless indicators for process, CPU, file, and local network activity.

```
(nimisha@Kali)-[~]  
$ mkdir -p ~/soc-incident-lab  
  
(nimisha@Kali)-[~]  
$ cd ~/soc-incident-lab
```

Figure 1 - SOC incident lab directory creation.

```
(nimisha@Kali)-[~/soc-incident-lab]  
$ logger -t SOC-LAB "Normal login user=kali source=192.168.56.10"  
logger -t SOC-LAB "System service started: NetworkManager"  
logger -t SOC-LAB "Normal file access: /home/kali/Documents/report.txt"  
logger -t SOC-LAB "User kali opened terminal"  
logger -t SOC-LAB "Normal DNS query to example.com"  
logger -t SOC-LAB "System update check completed"  
logger -t SOC-LAB "Normal SSH session closed user=kali"  
logger -t SOC-LAB "User kali created file notes.txt"  
logger -t SOC-LAB "Normal process started: firefox"  
logger -t SOC-LAB "Network connection established to 192.168.56.1"  
  
logger -t SOC-LAB "Failed login user=admin source=192.0.2.50"  
logger -t SOC-LAB "Failed login user=root source=192.0.2.50"  
logger -t SOC-LAB "Accepted login user=admin source=192.0.2.50"  
logger -t SOC-LAB "Suspicious hidden file created: /tmp/.soc-lab/update.sh"  
logger -t SOC-LAB "Suspicious process started: suspicious-updater"  
logger -t SOC-LAB "High CPU usage detected process=suspicious-updater cpu=92%"  
  
logger -t SOC-LAB "Normal process started: bash"  
logger -t SOC-LAB "User kali logged out normally"  
logger -t SOC-LAB "Security patch applied"  
logger -t SOC-LAB "System health check completed"
```

Figure 2 - Twenty mixed normal/suspicious SOC-LAB journal events, including failed logins, privileged login, hidden file, suspicious process, high CPU, patch, and normal activity.

2. Case 1 - Log Collection and Pre-Patch Evidence

Journal events were reviewed with ISO timestamps and exported to investigation_logs.txt. The suspicious authentication and activity entries appear before the patch event in log order.

```
(nimisha@Kali)-[~/soc-incident-lab]
$ journalctl -t SOC-LAB --no-pager -o short-iso
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8100]: Normal login user=kali source=192.168.56.10
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8101]: System service started: Network Manager
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8102]: Normal file access: /home/kali/Documents/report.txt
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8103]: User kali opened terminal
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8104]: Normal DNS query to example.com
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8105]: System update check completed
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8106]: Normal SSH session closed user=kali
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8107]: User kali created file notes.txt
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8108]: Normal process started: firefox
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8109]: Network connection established to 192.168.56.1
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8110]: Failed login user=admin source=192.0.2.50
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8111]: Failed login user=root source=192.0.2.50
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8112]: Accepted login user=admin source=192.0.2.50
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8113]: Suspicious hidden file created: /tmp/.soc-lab/update.sh
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8114]: Suspicious process started: suspicious-updater
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8115]: High CPU usage detected process=suspicious-updater cpu=92%
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8116]: Normal process started: bash
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8117]: User kali logged out normally
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8118]: Security patch applied
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8119]: System health check completed
```

Figure 3 - Journal review showing the full mixed event sequence.

```
(nimisha@Kali)-[~/soc-incident-lab]
$ journalctl -t SOC-LAB --no-pager -o short-iso > investigation_logs.txt

(nimisha@Kali)-[~/soc-incident-lab]
$ cat investigation_logs.txt
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8100]: Normal login user=kali source=192.168.56.10
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8101]: System service started: Network Manager
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8102]: Normal file access: /home/kali/Documents/report.txt
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8103]: User kali opened terminal
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8104]: Normal DNS query to example.com
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8105]: System update check completed
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8106]: Normal SSH session closed user=kali
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8107]: User kali created file notes.txt
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8108]: Normal process started: firefox
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8109]: Network connection established to 192.168.56.1
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8110]: Failed login user=admin source=192.0.2.50
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8111]: Failed login user=root source=192.0.2.50
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8112]: Accepted login user=admin source=192.0.2.50
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8113]: Suspicious hidden file created: /tmp/.soc-lab/update.sh
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8114]: Suspicious process started: suspicious-updater
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8115]: High CPU usage detected process=suspicious-updater cpu=92%
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8116]: Normal process started: bash
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8117]: User kali logged out normally
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8118]: Security patch applied
2026-09-10T17:01:04+05:30 Kali SOC-LAB[8119]: System health check completed
```

Figure 4 - Exported investigation_logs.txt preserving the same event sequence.

3. Case 1 - Process, CPU, File and Network Indicators

The suspicious indicators were intentionally harmless: a hidden test file, a renamed sleep process, a short-lived CPU load generator, and a localhost HTTP service on port 8080. These provide observable artifacts without using malware.

```
(nimisha@Kali)-[~/soc-incident-lab]
$ mkdir -p /tmp/.soc-lab

(nimisha@Kali)-[~/soc-incident-lab]
$ echo "SOC TEST FILE" > /tmp/.soc-lab/update.sh

(nimisha@Kali)-[~/soc-incident-lab]
$ bash -c 'exec -a suspicious-updater sleep 600' &
[1] 14448

(nimisha@Kali)-[~/soc-incident-lab]
$ timeout 20s yes > /dev/null &
[2] 14547

(nimisha@Kali)-[~/soc-incident-lab]
$ python3 -m http.server 8080 --bind 127.0.0.1
Serving HTTP on 127.0.0.1 port 8080 (http://127.0.0.1:8080/) ...
[2] + exit 124 timeout 20s yes > /dev/null
127.0.0.1 - - [10/Sep/2026 17:15:00] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [10/Sep/2026 17:17:51] "GET / HTTP/1.1" 200 -
[1] + done bash -c 'exec -a suspicious-updater sleep 600'
```

Figure 5 - Harmless suspicious-file, suspicious-process, short CPU-load, and localhost HTTP-service simulation.

```
(nimisha@Kali)-[~/soc-incident-lab]
$ curl http://127.0.0.1:8080
SOC-LAB[8110]: Network connection established
<!DOCTYPE HTML>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <title>Directory listing for /</title>
  </head>
  <body>
    <h1>Directory listing for /</h1>
    <hr>
    <ul>
      <li><a href="investigation_logs.txt">investigation_logs.txt</a></li>
    </ul>
  </body>
</html>
SOC-LAB[8110]: Failed login user=admin source=10.10.10.10
SOC-LAB[8111]: Failed login user=root source=10.10.10.10
SOC-LAB[8112]: Accepted login user=admin source=10.10.10.10
SOC-LAB[8113]: Suspicious hidden file created: /tmp/.soc-lab/update.sh
SOC-LAB[8114]: Suspicious process started: sleep 600
SOC-LAB[8115]: High CPU usage detected process: suspicious-updater
SOC-LAB[8116]: Normal process started: bash
SOC-LAB[8117]: User kali logged out normally
SOC-LAB[8118]: Security patch applied
SOC-LAB[8119]: System health check completed

(nimisha@Kali)-[~/soc-incident-lab]
$ logger -t SOC-LAB "Security patch applied"
```

Figure 6 - Local curl request to the test service and simulated patch event.

4. Automated Investigation - Case 1

The Python checker correlates authentication/log evidence with current process, CPU, file, and listening-port observations. It found multiple indicators and produced a HIGH RISK assessment.

```
Session Actions Edit View Help
(nimisha@Kali)-[~]
$ cd ~/soc-incident-lab
nano soc_auto_check.py lab

(nimisha@Kali)-[~/soc-incident-lab]
$ python3 soc_auto_check.py

===== AUTOMATED SOC INVESTIGATION =====
[1] AUTHENTICATION & LOG CHECK
[!] Failed login detected: 2026-09-10T17:01:04+05:30 Kali SOC-LAB[8110]: Failed login user=admin source=192.0.2.50
[!] Failed login detected: 2026-09-10T17:01:04+05:30 Kali SOC-LAB[8111]: Failed login user=root source=192.0.2.50
[!] Privileged successful login detected: 2026-09-10T17:01:04+05:30 Kali SOC-LAB[8112]: Accepted login user=admin source=192.0.2.50
[+] Patch event found: 2026-09-10T17:01:04+05:30 Kali SOC-LAB[8118]: Security patch applied
[!] Suspicious authentication activity occurred BEFORE patch

[2] PROCESS CHECK
[!] suspicious-updater process detected

[3] CPU CHECK
  PID  COMMAND  %CPU
  ---  ---
  17104 python3  28.5
  758 Xorg  1.2
  16693 qterminal  0.9
  1253 xfwm4  0.3
  1381 wrapper-2.0  0.3

[4] FILE CHECK
[!] Suspicious file found: /tmp/.soc-lab/update.sh

[5] NETWORK CHECK
[!] Unexpected/test service listening on port 8080

===== SOC VERDICT =====
HIGH RISK
Multiple suspicious indicators detected.
Evidence suggests suspicious activity existed before patching.
Further incident investigation is required.

Risk Score: 11
=====
```

Figure 7 - Automated Case 1 result: HIGH RISK, Risk Score 11.

5. Case 2 - Normal Activity Control

Case 2 was designed as the clean comparison: no simulated exploit indicators. A Host-Only adapter was used for local Windows-to-Kali SSH connectivity, and the user performed a normal successful login.

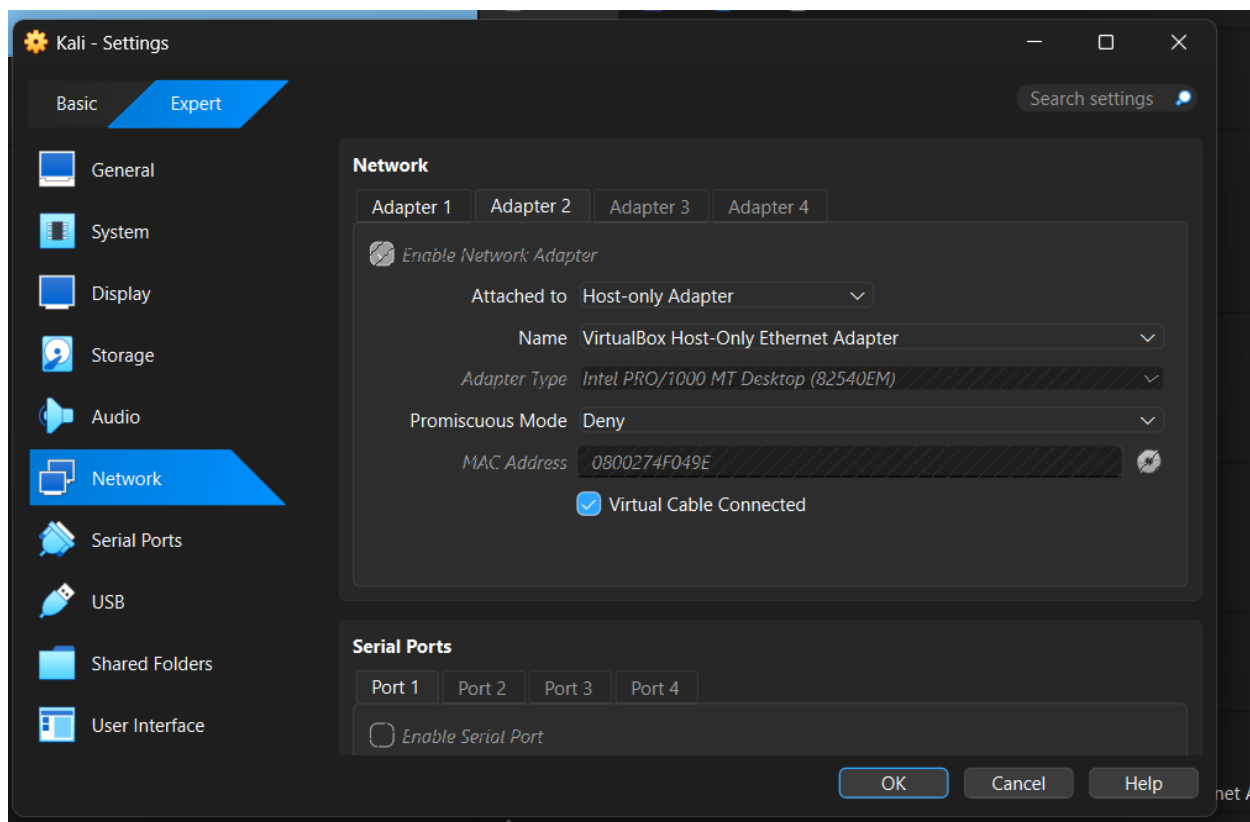


Figure 8 - VirtualBox Host-Only Adapter used for isolated host-to-VM communication.

```
C:\Users\LENOVO>ssh nimisha@192.168.56.103
The authenticity of host '192.168.56.103 (192.168.56.103)' can't be established.
ED25519 key fingerprint is SHA256:QY98xeNGyI6fCZjx8+nAHhH4T15Lyfn3yclXQpGv+f0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.103' (ED25519) to the list of known hosts.
nimisha@192.168.56.103's password:
Linux Kali 6.18.12+kali-amd64 #1 SMP PREEMPT_DYNAMIC Kali 6.18.12-1kali1 (2026-02-25) x86_64

The programs included with the Kali GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Kali GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
[nimisha@Kali]~$
```

Figure 9 - Successful normal SSH login from the Windows host to Kali at 192.168.56.103.

6. Case 2 - Collected Logs and Automated Result

The real SSH service generated successful authentication/session events. The Case 2 log set also contains the patch event, while Case 1 live indicators were removed before running the same automated checker.

```
(nimisha@Kali)-[~/soc-incident-lab]
$ logger -t SOC-CASE2 "Security patch applied"

(nimisha@Kali)-[~/soc-incident-lab]
$ journalctl --since "10 minutes ago" -o short-iso --no-pager | grep -Ei "ssh|SOC-CASE2" > case2_logs.txt

(nimisha@Kali)-[~/soc-incident-lab]
$ journalctl --since "15 minutes ago" -o short-iso --no-pager | grep -Ei "ssh|SOC-CASE2" > investigation_logs.txt

(nimisha@Kali)-[~/soc-incident-lab]
$ cat investigation_logs.txt
2026-09-10T17:38:41+05:30 Kali sshd-session[2997]: Accepted password for nimisha from 192.168.56.1 port 65249 ssh2
2026-09-10T17:38:41+05:30 Kali sshd-session[2997]: pam_unix(sshd:session): session opened for user nimisha(uid=1000) by nimisha(uid=0)
2026-09-10T17:40:10+05:30 Kali sshd-session[2997]: syslogin_perform_logout: logout() returned an error
2026-09-10T17:40:10+05:30 Kali sshd-session[3693]: Received disconnect from 192.168.56.1 port 65249:11: disconnected by user
2026-09-10T17:40:10+05:30 Kali sshd-session[3693]: Disconnected from user nimisha 192.168.56.1 port 65249
2026-09-10T17:40:10+05:30 Kali sshd-session[2997]: pam_unix(sshd:session): session closed for user nimisha
2026-09-10T17:40:24+05:30 Kali sshd-session[4624]: Accepted password for nimisha from 192.168.56.1 port 51139 ssh2
2026-09-10T17:40:24+05:30 Kali sshd-session[4624]: pam_unix(sshd:session): session opened for user nimisha(uid=1000) by nimisha(uid=0)
2026-09-10T17:42:34+05:30 Kali SOC-CASE2[5857]: Security patch applied
2026-09-10T17:50:33+05:30 Kali SOC-CASE2[9718]: Security patch applied
```

Figure 10 - Real SSH session events plus the Case 2 patch event collected into the investigation log.

```
(nimisha@Kali)-[~/soc-incident-lab]
$ python3 soc_auto_check.py

===== AUTOMATED SOC INVESTIGATION =====

[1] AUTHENTICATION & LOG CHECK
[+] Patch event found: 2026-09-10T17:42:34+05:30 Kali SOC-CASE2[5857]: Security patch applied
[+] Patch event found: 2026-09-10T17:50:33+05:30 Kali SOC-CASE2[9718]: Security patch applied

[2] PROCESS CHECK
[+] No suspicious-updater process detected

[3] CPU CHECK
    PID COMMAND      %CPU
    10016 python3      60.0
    694 Xorg           1.2
    1282 wrapper-2.0    0.3
    1154 xfwm4          0.3
    1280 wrapper-2.0    0.2

[4] FILE CHECK
[+] Suspicious test file not found

[5] NETWORK CHECK
[+] Port 8080 not detected

===== SOC VERDICT =====
LOW RISK
No strong evidence of compromise found.

Risk Score: 0
=====
```

Figure 11 - Automated Case 2 result: LOW RISK, Risk Score 0.

7. Investigation Analysis

Investigation Area	Case 1 Finding	Case 2 Finding
Identify / Patch	Patch event present after suspicious events	Patch event present
Authentication	Failed admin/root logins followed by accepted admin login	Admin successful SSH login
Process	suspicious-updater detected	No suspicious-updater detected
CPU	High-CPU activity was simulated; checker displayed top processes	No attack processes; checker only displayed current CPU
File	/tmp/.soc-lab/update.sh found	Test file absent
Persistence	Not directly simulated in this lab	No persistence indicator
Network	Local test service listening on 8080	Port 8080 absent
Correlation	Multiple indicators before patch	No strong compromise indicators
Verdict	HIGH RISK - further investigation required	LOW RISK - no strong evidence found

Timeline - Case 1

Because the dummy logger commands were executed very quickly, several events share the same second-level timestamp. Therefore, event order in the journal is used to show the sequence:



Key conclusion: A patch answers whether the vulnerability is fixed now. It does not prove that the endpoint was never compromised before remediation. Case 1 demonstrates this distinction; Case 2 demonstrates how the same checker behaves when no strong compromise indicators remain.

8. Detection Engineering

The Sigma rule below is intentionally simple and focuses on successful privileged authentication. In production, it should be correlated with preceding failures, source reputation, endpoint context, process/file creation, and network behavior.

```
title: Suspicious Privileged Login on Linux
id: 8b6f3b9a-2f0e-4e8a-9d11-111111111111
status: experimental
description: Detects successful login activity involving privileged users.

logsource:
  product: linux
  category: authentication

detection:
  selection:
    - 'Accepted login user=admin'
    - 'Accepted login user=root'
    - 'Accepted password for root'
  condition: selection

level: high
falsepositives:
  - Legitimate administrator login
  - Authorized maintenance activity

tags:
  - attack.initial-access
  - attack.t1078
```

Required Item	Implementation
Detection logic	Alert on unexpected successful privileged authentication; prioritize when preceded by failed attempts from
Data source	Linux authentication / SSH logs.
Relevant fields	Timestamp, username, source IP, authentication result, process/service.
Expected behavior	Known user from an expected source during legitimate activity.
Suspicious behavior	Unexpected admin/root success, especially after failures or followed by unusual endpoint activity.
Potential false positives	Authorized administrator access, maintenance, security testing.
Recommended analyst response	Validate user/source; review pre/post-login process, file, persistence and network activity; preserve evidence

MITRE ATT&CK; Mapping

T1078 - Valid Accounts can be used as a behavioral reference when an attacker uses a compromised or unauthorized valid account to authenticate. The lab's accepted admin-login event is simulated evidence and does not by itself prove T1078; correlation is required.

Recommended Response

Preserve relevant logs and timestamps; validate the account and source IP; review process tree, new/hidden files, persistence locations, and outbound connections; isolate the endpoint if evidence supports compromise; reset affected credentials; verify patch status; remove confirmed malicious artifacts; and continue monitoring.

Final Assessment

Case 1: HIGH RISK. Multiple simulated indicators appeared before the patch event; further investigation would be required in a real incident.

Case 2: LOW RISK. A normal successful SSH session was observed and the automated checker found no strong compromise artifacts. The correct wording is "no strong evidence of compromise found," not "the system was definitely never compromised."

9. Sigma Detection Rule - Case 1

This rule detects a successful privileged login on a Linux endpoint. In this lab, the alert is strengthened when the same source IP shows failed login attempts before the successful admin/root login.

```
title: Suspicious Privileged Login on Linux
id: 8b6f3b9a-2f0e-4e8a-9d11-111111111111
status: experimental
description: Detects successful privileged login activity for admin/root accounts.

logsource:
  product: linux
  category: authentication

detection:
  selection:
    message|contains:
      - 'Accepted login user=admin'
      - 'Accepted login user=root'
      - 'Accepted password for root'
  condition: selection

falsepositives:
  - Legitimate administrator login
  - Authorized maintenance activity

level: high

tags:
  - attack.initial-access
  - attack.t1078
```

Detection details

Detection logic	Alert on successful admin/root authentication; raise priority when preceded by failed attempts from the same source IP.
Data source	Linux authentication / SSH logs (journal, auth.log, or SIEM-normalized authentication events).
Relevant fields	Timestamp, username, source IP, authentication result, service/process.
Expected behavior	Known user authenticates from an expected source during legitimate activity.
Suspicious behavior	Unexpected privileged login, especially after repeated failures or from an unfamiliar source.
Potential false positives	Authorized admin maintenance, security testing, automation, or emergency access.
Analyst response	Validate user and source IP; review preceding failures; correlate process, file, persistence, and network activity; preserve evidence and isolate if confirmed malicious.

Correlation enhancement

In a SIEM, increase severity if the same source IP has 2 or more failed logins within 5 minutes before the privileged success.